Name: _____ **Date:** _____

Balancing Chemical Equations

Balance the following chemical equations.

1. ____ Na_3PO_4 + ____ HCl ⟶ ____ NaCl + ____ H_3PO_4

2. ____ Al + ____ HCl ⟶ ____ $AlCl_3$ + ____ H_2

3. ____ FeC_2O_4 + ____ CO_2 ⟶ ____ $Fe_2(C_2O_4)_3$

4. ____ $TiCl_4$ + ____ H_2O ⟶ ____ TiO_2 + ____ HCl

5. ____ SnO_2 + ____ H_2 ⟶ ____ Sn + ____ H_2O

6. ____ B_2Br_6 + ____ HNO_3 ⟶ ____ $B(NO_3)_3$ + ____ HBr

7. ____ C_2H_6O + ____ O_2 ⟶ ____ CO_2 + ____ H_2O

8. ____ Zn + ____ HCl ⟶ ____ $ZnCl_2$ + ____ H_2

9. ____ As + ____ NaOH ⟶ ____ Na_3AsO_3 + ____ H_2

10. ____ $SeCl_6$ + ____ O_2 ⟶ ____ SeO_2 + ____ Cl_2

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Answer key

1. $Na_3PO_4 + 3KOH \rightarrow 3NaOH + K_3PO_4$
2. $MgF_2 + Li_2CO_3 \rightarrow MgCO_3 + 2LiF$
3. $P_4 + 3O_2 \rightarrow 2P_2O_3$
4. $RbNO_3 + BeF_2 \rightarrow Be(NO_3)_2 + 2RbF$
5. $2AgNO_3 + Cu \rightarrow Cu(NO_3)_2 + 2Ag$
6. $CF_4 + 2Br_2 \rightarrow CBr_4 + 2F_2$
7. $2HCN + CuSO_4 \rightarrow H_2SO_4 + Cu(CN)_2$
8. $GaF_3 + 3Cs \rightarrow 3CsF + Ga$
9. $BaS + PtF_2 \rightarrow BaF_2 + PtS$
10. $N_2 + 3H_2 \rightarrow 2NH_3$
11. $2NaF + Br_2 \rightarrow 2NaBr + F_2$
12. $Pb(OH)_2 + 2HCl \rightarrow 2H_2O + PbCl_2$
13. $2AlBr_3 + 3K_2SO_4 \rightarrow 6KBr + Al_2(SO_4)_3$
14. $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$
15. $2Na_3PO_4 + 3CaCl_2 \rightarrow 6NaCl + Ca_3(PO_4)_2$
16. $2K + Cl_2 \rightarrow 2KCl$
17. $2Al + 6HCl \rightarrow 3H_2 + 2AlCl_3$
18. $N_2 + 3F_2 \rightarrow 2NF_3$
19. $SO_2 + Li_2Se \rightarrow SSe_2 + Li_2O$
20. $2NH_3 + H_2SO_4 \rightarrow (NH_4)_2SO_4$

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